Select Problems in Stochastic Calculus

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May 14, 2026

1 Prove that every stationary Gaussian Markov process is an Ornstein-Uhlenbeck process.

We will approach this problem by using a result from Basu's textbook:

Lemma: A zero-mean Gaussian process has the Markov property iff its covariance satisfies $C(s,u)C(t,t) = C(s,t)C(t,u)$ when $s < t < u$.

Proof:
Forward direction: Assume $X_t \sim \mathcal{N}(0, C)$ is Markov with respect to the filtration $\mathcal{F}_t := \sigma(X_s, s \leq t)$. For Gaussian random variables X, Y,

$$\mathbb{E}[X|Y] = \mathbb{E}[X] + \Sigma_{XY}\Sigma_{YY}^{-1}(Y - \mathbb{E}[Y])$$

where Σ is the covariance matrix between X, Y. So we get

$$\mathbb{E}[X_u|\mathcal{F}_s] = \mathbb{E}[X_u|X_s] = \frac{C(s,u)}{C(s,s)}X_s$$

But also, by the tower law,

$$\begin{aligned}
\mathbb{E}[X_u|\mathcal{F}_s] &= \mathbb{E}[\mathbb{E}[X_u|\mathcal{F}_t]|\mathcal{F}_s] \\
&= \mathbb{E}[\frac{C(t,u)}{C(t,t)}X_t|\mathcal{F}_s] \\
&= \frac{C(t,u)C(s,t)}{C(t,t)C(s,s)}X_s
\end{aligned}$$

yielding $C(s,u)C(t,t) = C(s,t)C(t,u)$ as desired.

Backward direction: Assume $C(s,u)C(t,t) = C(s,t)C(t,u)$ holds $\forall s < t < u$. By the nature of conditional expectation, $X_u - \mathbb{E}[X_u|X_t] \perp X_t$, so

$$X_u - \frac{C(t,u)}{C(t,t)}X_t \perp X_t$$

Computing the covariance between X_s and $X_u - \frac{C(t,u)}{C(t,t)}X_t$ for $s < t$,

$$\mathbb{E}\left[(X_u - \frac{C(t,u)}{C(t,t)}X_t)X_s\right] = C(s,u) - \frac{C(t,u)C(s,t)}{C(t,t)} = 0$$

The Gaussians are uncorrelated so $X_u - \mathbb{E}[X_u|X_t] \perp X_s \forall s < t$. It remains to show $\mathbb{E}[f(X_u)|\mathbb{F}_t] = \mathbb{E}[f(X_u)|X_t]$ which is equivalent to the Markov property if it holds for all bounded, measurable f.

$$\begin{aligned}
\mathbb{E}[f(X_u)|\mathcal{F}_t] &= \mathbb{E}[f(\mathbb{E}[X_u|X_t] + X_u - \mathbb{E}[X_u|X_t]|\mathcal{F}_t] \\
&= \mathbb{E}[f(Z+y)]_{y=\mathbb{E}[X_u|X_t]}
\end{aligned}$$

by letting $Y := \mathbb{E}[X_u|X_t], Z := X_u - \mathbb{E}[X_u|X_t]$. Note that Y is measurable with respect to $\mathcal{F}_t$ and Z is independent of it, so by the substitution lemma, $\mathbb{E}[f(Z+Y)]$ conditions only on X_t which holds $\forall t \leq u$, which gives X_t its Markov property.

To be specific, $Z \perp \mathcal{F}_t$ arises from $\text{Cov}(X_u - \mathbb{E}[X_u|X_t], X_s) = 0 \ \forall s \leq t$ so Z is independent of all linear combinations of X_s as well, and $\mathcal{F}_t$ is spanned by the closure of linear combinations of Gaussians X_s. $\square$

Back to the actual problem, an Ornstein-Uhlenbeck process is a stationary process with mean μ and covariance $C(s,t) = \frac{\sigma^2}{2\theta}e^{-\theta|t-s|}$ governed by the SDE

$$dX_t = \theta(\mu - X_t)dt + \sigma dW_t, \theta > 0$$

Suppose X_t is Gaussian, stationary and Markov. By the lemma, $C(s,u)C(t,t) = C(s,t)C(t,u)$, and by stationarity, $C(|u-s|)C(0) = C(|t-s|)C(|u-t|)$ for $s < t < u$. Let $x := |t-s|$, $y := |u-t|$, $h(x) := \frac{C(x)}{C(0)}$. Then,

$$C(x+y)C(0) = C(x)C(y)$$

and dividing by $C(0)^2$ on each side,

$$h(x+y) = h(x)h(y)$$

Since h is an adaptation of a Gaussian covariance function, it is continuous and satisfies Cauchy's exponential functional equation, which has solutions $h(x) = 0$ or $h(x) = e^{cx}$. Observe that we get the equations

$$\begin{aligned} h(x) &= h(1+1+...1) = h(1)^x & x \in \mathbb{Z} \\ h(\frac{p}{q}) &= h(\frac{1}{q} + ...\frac{1}{q}) = h(\frac{1}{q})^p & x \in \mathbb{Q} \end{aligned}$$

Since $h(1) = h(\frac{1}{q})^q, h(\frac{p}{q}) = h(1)^{\frac{p}{q}}$. By density of $\mathbb{Q}$ and continuity, $h(x) = h(1)^x = e^{x \ln h(1)} \ \forall x > 0$, so

$$C(x) = C(0)e^{x \ln\left(\frac{C(1)}{C(0)}\right)}$$

Recall that we have defined x as a distance between times due to stationarity of the process, so $C(x) = C(-x) = C(|t-s|)$ for $s,t \in \mathbb{R}$. The equation becomes

$$C(x) = C(0)e^{|x| \ln\left(\frac{C(1)}{C(0)}\right)}$$

Let $t = s+1$, so $C(1) = C(s,t), C(0) = C(s,s)$. Since $|\rho| := \frac{|C(s,t)|}{\sqrt{C(s,s)C(t,t)}} \leq 1$, $C(1) \leq C(0)$ so $\ln\left(\frac{C(1)}{C(0)}\right) \leq 0$. Define $\theta := -\ln\frac{C(1)}{C(0)}$, $\sigma^2 := 2\theta C(0)$ to give

$$C(s,t) = C(|t-s|) = \frac{\sigma^2}{2\theta}e^{-\theta|t-s|}$$

which is precisely the covariance function of the O-U process.

To deal with the mean μ, since the process is stationary, μ is constant for all t, so we can consider the zero-mean $Y_t = X_t - \mu$, which is a O-U process. Then, adding the constant mean back just yields a shifted O-U SDE, as the covariance structure doesn't change if we subtract a mean. Since these are Gaussians, they are uniquely determined by μ, C. Because the covariances match, effectively all stationary Gaussian Markov processes are O-U processes. $\square$

2 Compute the distribution of

$$\int_0^\tau f(s)\mathbf{d}B(s)$$

under the conditions: $B(t)$ is a Brownian motion, f is progressively measurable, $\mathbb{E}[\int_0^t f^2(s)\mathbf{d}s] < \infty$, $\int_0^\infty f^2(s)\mathbf{d}s = \infty$ almost surely, and the stopping time is defined as:

$$\tau := \inf\left\{t \geq 0 : \int_0^t f^2(s)\mathbf{d}s = \sigma^2\right\}$$

This problem can be solved using a relatively direct application of the Dubins-Schwarz theorem, but first we need to set up some things.

Consider the martingale $M_t = \int_0^t f(s)dB_s$; we will show that its quadratic variation, defined as $\lim_{|\Pi|\downarrow 0} \sum_{i=1}^n (M_{t_i} - M_{t_{i-1}})^2$ is $\int_0^t f^2 ds$ via convergence in probability.

$$\begin{aligned}
\mathbb{P}\left(\left|\sum_{i=1}^n (M_{t_i} - M_{t_{i-1}})^2 - \int_0^t f^2(s)\mathrm{d}s\right| > \epsilon\right) &= \mathbb{P}\left(\left|\sum_{i=1}^n \left(\int_{t_{i-1}}^{t_i} f(s)\mathrm{d}B_s\right)^2 - \int_0^t f^2(s)\mathrm{d}s\right| > \epsilon\right) \\
&\leq \frac{1}{\epsilon^2}\mathrm{Var}\left(\sum_{i=1}^n \left(\int_{t_{i-1}}^{t_i} f(s)\mathrm{d}B_s\right)^2\right) \\
&= \frac{1}{\epsilon^2}\sum_{i=1}^n \left(\mathrm{Var}\left(\int_{t_{i-1}}^{t_i} f(s)\mathrm{d}B_s\right)^2\right) \\
&= \frac{1}{\epsilon^2}\sum_{i=1}^n \left(3\mathrm{Var}\left(\int f\mathrm{d}B\right)^2 - \mathrm{Var}\left(\int f\mathrm{d}B\right)^2\right) \\
&= \frac{2}{\epsilon^2}\sum_{i=1}^n \left(\int_{t_{i-1}}^{t_i} f^2(s)\mathrm{d}s\right)^2
\end{aligned}$$

where the third step is justified by independence of disjoint Brownian increments, the fourth step is justified by Isserlis-Wick identity, and the fifth step follows from Ito's isometry.

Since $f \in L^2$ on finite time domains,

$$\sum_{i=1}^n \left(\int_{t_{i-1}}^{t_i} f^2(s)\mathrm{d}s\right)^2 \leq \sup_n \left[\int_{t_{i-1}}^{t_i} f^2(s)\mathrm{d}s\right] \int_0^t f^2(s)\mathrm{d}s$$

with the square integral being uniformly continuous on a compact set and thus converging to 0 as $n \to \infty$. Thus, the quadratic variation of M_t at infinity, $QV(M_t)_\infty = \infty$ by assumption.

Since $\int_0^t f^2\mathrm{d}s$ is continuous and non-decreasing, the stopping time τ is equivalent to

$$\tau = \inf\left\{t \geq 0 : \int_0^t f^2(s,\omega)\mathrm{d}s > \sigma^2\right\}$$

Now we are ready to make use of the theorem.

Lemma (Dubins-Schwarz): Let M_t be a $\mathcal{F}_t$-adapted continuous local martingale (with respect to the usual filtration), and its quadratic variation $QV(M)_\infty = \infty$, $M_0 = 0$. Define $\forall t \geq 0$, $\tau_t := \inf\{s : QV(M)_s > t\}$. Then M_{τ_t} is a $\mathcal{F}_{\tau_t}$-adapted Brownian motion, with $M_t = B_{QV(M)_t}$.

Proof:
We seek to show that M_{τ_t} is a continuous martingale with quadratic variation equal to t, using the Levy characterization of Brownian motion.

Let $W_s := M_{\tau_s}$ be adapted to $\mathcal{F}_{\tau_s}$, the filtration of stopping times $\tau_s := \inf\left\{t \geq 0 : \int_0^t f^2 > s\right\}$. Clearly, $s \mapsto \tau_s$ is continuous and non-decreasing. Observe that if $\int_0^t f^2 \mathrm{d}u$ is constant on $[t_1, t_2]$, then $QV(M)_{t_2} - QV(M)_{t_1} = 0$. Since $\int_0^t f^2 \mathrm{d}u$ is constant, $\int_0^t f \mathrm{d}B_u$ is constant almost surely, which is a characteristic of martingales–thus $M_{t_1} = M_{t_2}$ almost surely. Also, M_{τ_s} is continuous because it is a composition of M, τ, both continuous functions.

We also know that M_{τ_s} is a martingale on $\mathcal{F}_{\tau_s}$ by optional stopping. Furthermore,

$$QV(M)_{\tau_s} = \int_0^{\tau_s} f^2 \mathrm{d}u = s$$

by definition, so M_{τ_s} is a Brownian motion B_s. To conclude the final result,

$$B_{QV(M)_t} = M_{\tau_{QV(M)_t}}$$

almost surely $\forall t$, and $\tau_{QV(M)_t} = t$ because intuitively, τ_s and $QV(M)_t$ are continuous and thus takes the same value "near" the stopping time. $\square$

Using the theorem, $M_\tau = \int_0^\tau f \mathrm{d}B_s$ is a Brownian motion, and it is easy to see from definitions that

$$\int_0^\tau f \mathrm{d}B_s = M_\tau = B_{QV(M)_\tau} = B_{\int_0^\tau f^2 \mathrm{d}s} = B_{\sigma^2}$$

Thus,

$$\int_0^\tau f(s,\omega)\mathrm{d}B_s \sim \mathcal{N}(0, \sigma^2)$$

3 Let $B(t) = (B_1(t), B_2(t))$ be the two-dimensional Brownian bridge on [0,1] from (0,0) to (0,0), i.e. B_1, B_2 are two independent Brownian bridges on [0,1]–the zero-mean Gaussian process with covariance $\mathbb{E}[B(t)B(s)] = s \wedge t - st$. The modified Levy's stochastic area is defined as:

$$A := \frac{1}{2}\int_0^1 B_1 \mathbf{d}B_2 - B_2 \mathbf{d}B_1 = \int_0^1 B_1 \mathbf{d}B_2 = \lim_{n\to\infty} A_n$$
$$A_n = \sum_{k=0}^{n-1} B_1(\frac{k}{n})(B_2(\frac{k+1}{n}) - B_2(\frac{k}{n}))$$

1. **Prove the limit characterization in the identity**
2. **Compute the characteristic function of the Levy area**

Part 1: We see to show that

$$\lim_{n\to\infty} \sum_{k=0}^{n-1} B_1(\frac{k}{n})(B_2(\frac{k+1}{n}) - B_2(\frac{k}{n})) = \int_0^1 B_1 \mathrm{d}B_2$$

where

$$B_1(t) = W_1(t) - tW_1(1)$$
$$B_2(t) = W_2(t) - tW_2(1)$$

with W_1, W_2 independent Brownian motions. Expand the identity:

$$\lim_{n\to\infty}\sum_{k=0}^{n-1}\left(W_1\left(\frac{k}{n}\right)-\frac{k}{n}W_1(1)\right)\left(W_2\left(\frac{k+1}{n}\right)-\frac{k+1}{n}W_2(1)-W_2\left(\frac{k}{n}\right)+\frac{k}{n}W_2(1)\right)$$
$$=\lim_{n\to\infty}\sum_{k=0}^{n-1}\underbrace{W_1\left(\frac{k}{n}\right)\left(W_2\left(\frac{k+1}{n}\right)-W_2\left(\frac{k}{n}\right)\right)}_{(I)}\underbrace{-\frac{W_2(1)}{n}\sum_{k=0}^{n-1}W_1\left(\frac{k}{n}\right)}_{(II)}$$
$$\underbrace{-\frac{W_1(1)}{n}\sum_{k=0}^{n-1}k\left(W_2\left(\frac{k+1}{n}\right)-W_2\left(\frac{k}{n}\right)\right)}_{(III)}\underbrace{+\frac{W_1(1)W_2(1)}{n^2}\sum_{k=0}^{n-1}k}_{(IV)}$$

As $n\to\infty$,

$$(I)\to\int_0^1 W_1(s)\mathrm{d}W_2(s)$$
$$(II)\to -W_2(1)\int_0^1 W_1(s)\mathrm{d}s$$
$$(III)\to -W_1(1)\int_0^1 s\mathrm{d}W_2(s)$$
$$(IV)\to\frac{W_1(1)W_2(1)}{2}$$

Applying integration by parts to (III),

$$\lim_{n\to\infty}A_n=\int_0^1 W_1(s)\mathrm{d}W_2(s)-W_2(1)\int_0^1 W_1(s)\mathrm{d}s-W_1(1)\left[W_2(1)-\int_0^1 W_2(s)\mathrm{d}s\right]+\frac{W_1(1)W_2(1)}{2}$$

Ito's lemma on B_2 gives us $dB_2(s)=dW_2(s)-W_2(1)ds$, so

$$\int_0^1 B_1\mathrm{d}B_2=\int_0^1 W_1(s)\mathrm{d}B_2(s)-\int_0^1 sW_1(1)\mathrm{d}B_2(s)$$
$$=\int_0^1 W_1(s)\mathrm{d}W_2(s)-W_2(1)\int_0^1 W_1(s)\mathrm{d}s-W_1(1)\int_0^1 s\mathrm{d}W_2(s)-W_1(1)W_2(1)\underbrace{\int_0^1 s\mathrm{d}s}_{=\frac{1}{2}}$$

Since the two equations above are equal, $\lim_{n\to\infty}A_n=\int_0^1 B_1\mathrm{d}B_2$ as desired. $\square$

Part 2: We seek to compute $\mathbb{E}[e^{i\lambda A}]$, which we will first do by evaluating $\mathbb{E}[e^{i\lambda A_n}]=\mathbb{E}[\mathbb{E}[e^{i\lambda A_n}|B_1]]$ and justifying taking a limit. Consider $A_n|B_1$, where the path of B_1 is deterministic:

$$A_n|B_1=\sum_{k=0}^{n-1}\underbrace{c_k}_{B_1\left(\frac{k}{n}\right)}\left(B_2\left(\frac{k+1}{n}\right)-B_2\left(\frac{k}{n}\right)\right)$$

Clearly, $\mathbb{E}[A_n|B_1]=0$, so this is a zero-mean Gaussian. To evaluate its variance, we need the covariance structure of Brownian bridge increments: $\mathbb{E}[B_2(s)B_2(t)]=s\wedge t-st$. Let $\Delta B_2^k:=B_2\left(\frac{k+1}{n}\right)-B_2\left(\frac{k}{n}\right)$. Then,

$$\mathbb{E}[\Delta B_2^k\Delta B_2^j]=\mathbb{E}\left[B_2\left(\frac{k+1}{n}\right)B_2\left(\frac{j+1}{n}\right)-B_2\left(\frac{k}{n}\right)B_2\left(\frac{j+1}{n}\right)-B_2\left(\frac{j}{n}\right)B_2\left(\frac{k+1}{n}\right)+B_2\left(\frac{k}{n}\right)B_2\left(\frac{j}{n}\right)\right]$$

Proceed by exhausting all integer cases of j,k. Assume $j<k$, so $j\wedge k=j$, $(j+1)\wedge k=j+1$. WLOG, the $k<j$ case is equivalent. Then assume $j=k$ in which case the covariance becomes $\mathrm{Var}(\Delta B_2^k)$. Computing

via the covariance structure gives us

$$\mathbb{E}[\Delta B_2^k \Delta B_2^j] = \begin{cases} -\frac{1}{n^2} & j \neq k \\ \frac{1}{n} - \frac{1}{n^2} & j = k \end{cases}$$

Thus, the increments $B_2\left(\frac{k+1}{n}\right) - B_2\left(\frac{k}{n}\right) \sim \mathcal{N}(0, \Sigma)$, where $\Sigma_{ij} = \mathbb{E}[\Delta B_2^i \Delta B_2^j]$.

So $A_n | B_1 \sim \mathcal{N}(0, \sigma^2)$ as it is a sum of Gaussians, where

$$\begin{aligned} \sigma^2 &= \sum_{k=0}^{n-1} c_k^2 \mathrm{Var}(\Delta B_2^k) + \sum_{k \neq j}^{n-1} c_k c_j \mathbb{E}[\Delta B_2^i \Delta B_2^j] \\ &= \left(\frac{1}{n} - \frac{1}{n^2}\right) \sum_k c_k^2 - \frac{1}{n^2} \sum_{k \neq j} c_k c_j \\ &= \left(\frac{1}{n} - \frac{1}{n^2}\right) \sum_k c_k^2 - \frac{1}{n^2} \left(\left(\sum_k c_k\right)^2 - \sum_k c_k^2\right) \\ &= \frac{1}{n} \sum_k c_k^2 - \frac{1}{n^2} \left(\sum_k c_k\right)^2 \end{aligned}$$

Also, the characteristic function of a zero-mean gaussian is given by $\phi(\lambda) = \mathbb{E}[e^{i\lambda x}] = e^{-\frac{\sigma^2 \lambda^2}{2}}$, so

$$\mathbb{E}[e^{i\lambda A_n} | B_1] = e^{-\frac{\lambda^2}{2}\left(\frac{1}{n}\sum_k B_1\left(\frac{k}{n}\right)^2 - \frac{1}{n^2}\left(\sum_k B_1\left(\frac{k}{n}\right)\right)^2\right)}$$

Now we must take

$$\lim_{n \to \infty} \mathbb{E}\left[e^{-\frac{\lambda^2}{2}\left(\frac{1}{n}\sum_k B_1\left(\frac{k}{n}\right)^2 - \frac{1}{n^2}\left(\sum_k B_1\left(\frac{k}{n}\right)\right)^2\right)}\right]$$

By bounded convergence theorem on the exponential, we can swap the expectation and limit. As $n \to \infty$,

$$\frac{1}{n} \sum_k B_1^2(\frac{k}{n}) \to \int_0^1 B_1^2(s) \mathrm{d}s$$

$$-\frac{1}{n^2} \left(\sum_k B_1\left(\frac{k}{n}\right)\right)^2 \to -\left(\int_0^1 B_1(s) \mathrm{d}s\right)^2$$

And we get

$$\mathbb{E}[e^{i\lambda A}] = \mathbb{E}\left[e^{-\frac{\lambda^2}{2}\left(\int_0^1 B_1^2(s)\mathrm{d}s - \left(\int_0^1 B_1(s)\mathrm{d}s\right)^2\right)}\right]$$

Define $W := \int_0^1 B_1^2 ds - (\int_0^1 B_1 ds)^2$. To simplify this expression, we appeal to the Karhunen-Loeve expansion of the Brownian bridge,

$$B_1(s) = \sum_{k=1}^{\infty} \frac{\sqrt{2}}{\pi k} \sin(\pi k s) \xi_k$$

where $\xi_k \sim \mathcal{N}(0,1)$ i.i.d. across k. Recall that $\sqrt{2}\sin(\pi k s)$ are orthonormal in $L^2([0,1])$ with respect to the standard inner product, so the integrals in W simplify to:

$$\int_0^1 B_1^2(s)\mathrm{d}s = \sum_{k=1}^{\infty} \frac{1}{k^2\pi^2}\xi_k^2$$
$$\int_0^1 B_1(s)\mathrm{d}s = \sum_{k=1}^{\infty} \xi_k \underbrace{\int_0^1 \frac{\sqrt{2}}{k\pi}\sin(k\pi s)\mathrm{d}s}_{(\gamma_k)}$$
$$\gamma_k := \frac{\sqrt{2}}{k^2\pi^2}(1-(-1)^k) = \begin{cases} \frac{2\sqrt{2}}{k^2\pi^2} & k \text{ odd} \\ 0 & k \text{ even} \end{cases}$$

Now that W is a difference of sums, we can rewrite it as a quadratic form:

$$W = \Xi^T(D - \Gamma\Gamma^T)\Xi$$
$$D := \operatorname{diag}\left(\frac{1}{k^2\pi^2}\right) = \operatorname{diag}((d_k)_k), d_k := \frac{1}{k^2\pi^2}$$
$$\Gamma := (\gamma_k)_k$$
$$\Xi := (\xi_k)_k$$

This is in particular, a Gaussian quadratic form of standard Gaussians, whose characteristic function can be found in Schwartz, Bennett, and Stein's "Communication Systems and Techniques":

$$\mathbb{E}\left[e^{-\frac{\lambda^2}{2}\Xi^T(D-\Gamma\Gamma^T)\Xi}\right] = \left(\sqrt{\det(I+\lambda^2(D-\Gamma\Gamma^T))}\right)^{-1}$$
$$= \left(\sqrt{\det(I+\lambda^2(D-\Gamma\Gamma^T)_{even})\det(I+\lambda^2(D-\Gamma\Gamma^T)_{odd}}\right)^{-1}$$

where the second equation is obtained by re-ordering the indices of the matrix to group odd indices together at one end, and even indices at the other, leading to block diagonal structure of $(D-\Gamma\Gamma^T)$ (because for even indices, $\Gamma_i = 0$). For $(D-\Gamma\Gamma^T)_{even} = D$,

$$\det(I+\lambda^2(D-\Gamma\Gamma^T)_{even}) = \prod_k \left(1 + \frac{\lambda^2}{4k^2\pi^2}\right)$$
$$= \frac{\sinh(\lambda/2)}{\lambda/2}$$

by Euler's sinh product expansion. For $(D-\Gamma\Gamma^T)_{odd}$, apply the matrix determinant lemma:

$$\det(I+\lambda^2(D_{odd}-\Gamma_{odd}\Gamma_{odd}^T)) = \underbrace{\det(I+\lambda^2 D_{odd})}_{(*)}\underbrace{(1-\lambda^2\Gamma_{odd}^T(I+\lambda^2 D_{odd})^{-1}\Gamma_{odd})}_{(**)}$$
$$(*) = \prod_k \left(1+\frac{\lambda^2}{(2k+1)^2\pi^2}\right) = \cosh\left(\frac{\lambda}{2}\right)$$
$$(**) = 1 - \lambda^2 \sum_{k=0}^{\infty} \frac{\gamma_{2k+1}^2}{1+\lambda^2 d_{2k+1}}$$
$$= 1 - \lambda^2 \sum_{k=0}^{\infty} \frac{\frac{8}{(2k+1)^4\pi^4}}{1+\frac{\lambda^2}{(2k+1)^2\pi^2}}$$
$$= 1 - 8\sum_{k=0}^{\infty}\left(\frac{1}{(2k+1)^2\pi^2} - \frac{1}{(2k+1)^2\pi^2+\lambda^2}\right)$$
$$= 1 - 8 \underbrace{\frac{\pi^2}{8\pi^2}}_{\text{Basel sum}} + \underbrace{\frac{2}{\lambda}\tanh(\frac{\lambda}{2})}_{\text{partial fraction expansion of tanh}}$$

Putting all the terms together, we get

$$\begin{aligned}\mathbb{E}\left[e^{-\frac{\lambda^2}{2}\Xi^T(D-\Gamma\Gamma^T)\Xi}\right] &= \left(\sqrt{\frac{\sinh(\lambda/2)}{\lambda/2}\cosh\left(\frac{\lambda}{2}\right)\frac{2}{\lambda}\tanh(\frac{\lambda}{2})}\right)^{-1} \\ &= \frac{\lambda/2}{\sinh(\lambda/2)}\end{aligned}$$

which is the desired characteristic function, and in agreement with Levy's result in "Wiener's Random Function, and other Laplacian random functions" (1950) of Levy Area when conditioned on $(B_1(1), B_2(1)) = (0, 0)$.

4 Prove pathwise uniqueness for solutions of

$$dX_t = \sigma(t, X_t)dB_t + b(t, X_t)dt, X_0 = x_0$$

with the weakened condition of σ being α-Holder, $\frac{1}{2} < \alpha < 1$ in x (second argument), and b Lipschitz.

We begin the proof by considering two solutions, X_t^1, X_t^2 of the same SDE, and seek to show $\mathbb{E}[|X_t^1 - X_t^2|]$ vanishes. To approximate $|\cdot|$, we appeal to the family of functions $\phi_\epsilon(x) = (\epsilon + x^2)^{\frac{1}{2}}$, which uniformly converges to $|x|$.

Applying Ito's lemma to $\phi_\epsilon(X_t^1 - X_t^2)$, we get:

$$\begin{aligned}d\phi_\epsilon(X_t^1 - X_t^2) &= \phi'_\epsilon(X_t^1 - X_t^2)d(X_t^1 - X_t^2) + \frac{1}{2}\phi''_\epsilon(X_t^1 - X_t^2)d\underbrace{QV(X_t^1 - X_t^2)}_{\text{Quadratic Variation}} \\ \phi'_\epsilon(x) &= \frac{x}{\sqrt{\epsilon + x^2}} \\ \phi''_\epsilon(x) &= \frac{\epsilon}{(\epsilon + x^2)^{\frac{3}{2}}}\end{aligned}$$

where the quadratic variation is given by

$$\begin{aligned}dQV(X_t^1 - X_t^2) &= \underbrace{dQV(X_t^1)}_{\sigma(t,X_t^1)dt} + \underbrace{dQV(X_t^2)}_{\sigma(t,X_t^2)dt} - 2d\underbrace{CV(X_t^1, X_t^2)}_{\text{Cross Variation}} \\ dCV(X_t^1, X_t^2) &= \sigma(t, X_t^1)\sigma(t, X_t^2)dt\end{aligned}$$

where dCV becomes the product as we consider the same Brownian differential along the path of both solutions. We can thus rewrite $dQV(X_t^1 - X_t^2) = (\sigma(t, X_t^1) - \sigma(t, X_t^2))^2$.

Putting the Ito differential equation in integral form,

$$\begin{aligned}\phi_\epsilon(X_t^1 - X_t^2) = \phi_\epsilon(X_0^1 - X_0^2) &+ \int_0^t \frac{(X_s^1 - X_s^2)}{\sqrt{\epsilon + (X_s^1 - X_s^2)^2}}(b(s, X_s^1) - b(s, X_s^2))\mathrm{d}s \\ &+ \int_0^t \frac{(X_s^1 - X_s^2)}{\sqrt{\epsilon + (X_s^1 - X_s^2)^2}}(\sigma(s, X_s^1) - \sigma(s, X_s^2))\mathrm{d}B_s \\ &+ \frac{1}{2}\int_0^t \frac{\epsilon}{(\epsilon + (X_s^1 - X_s^2)^2)^{\frac{3}{2}}}(\sigma(s, X_s^1) - \sigma(s, X_s^2))^2\mathrm{d}s\end{aligned}$$

Define $\Delta_t := X_t^1 - X_t^2$ for convenience, and taking expectation of the above, we get

$$\mathbb{E}[\phi_\epsilon(\Delta_t)] = \mathbb{E}[\phi_\epsilon(\Delta_0)] + \underbrace{\mathbb{E}\left[\int_0^t \frac{\Delta_s}{\sqrt{\epsilon+\Delta_s^2}}(b(s,X_s^1) - b(s,X_s^2))\mathrm{d}s\right]}_{(I)}$$
$$+ \underbrace{\mathbb{E}\left[\int_0^t \frac{\Delta_s}{\sqrt{\epsilon+\Delta_s^2}}(\sigma(s,X_s^1) - \sigma(s,X_s^2))\mathrm{d}B_s\right]}_{(II)}$$
$$+ \underbrace{\frac{1}{2}\mathbb{E}\left[\int_0^t \frac{\epsilon(\sigma(s,X_s^1) - \sigma(s,X_s^2))^2}{(\epsilon+\Delta_s^2)^{\frac{3}{2}}}\mathrm{d}s\right]}_{(III)}$$

We can justify (II) vanishing over expectation, because $\frac{\Delta_s}{\sqrt{\epsilon+\Delta_s^2}} \leq 1$, and $|\sigma(s,X_s^1) - \sigma(s,X_s^2)| \leq C|\Delta_s|^\alpha$, so the integrand is L^2 for $\alpha \leq 1$ if $|\Delta_s| \in L^2$, and thus we are taking expectation over an Ito integral which is a martingale. Now, to bound the other two terms,

$$(I) \leq \mathbb{E}\left[\int_0^t C_1|\Delta_s|\mathrm{d}s\right]$$
$$(III) \leq \mathbb{E}\left[\int_0^t \frac{C_2|\Delta_s|^{2\alpha}\epsilon}{(\epsilon+\Delta_s^2)^{\frac{3}{2}}}\mathrm{d}s\right]$$

where the first inequality comes straight from Lipschitz condition on b, and the second inequality from Holder on σ. It would be very convenient to get (III)'s bound in terms of a linear $|\Delta_s|$ on the right hand side (like (I)'s bound) or have it vanish. Since we already have ϵ in the numerator, we should examine the convergence of the integrand $\frac{\epsilon|\Delta_s|^{2\alpha}}{(\epsilon+\Delta_s^2)^{\frac{3}{2}}}$:

First, the case when $\Delta_s^2 \geq \epsilon$:

$$(\epsilon+\Delta_s^2)^{\frac{3}{2}} \geq |\Delta_s|^3 \implies \frac{\epsilon|\Delta_s|^{2\alpha}}{(\epsilon+\Delta_s^2)^{\frac{3}{2}}} \leq \epsilon|\Delta_s|^{2\alpha-3} \leq \epsilon^{\alpha-\frac{1}{2}}$$

with the last inequality arising because $\alpha \leq 1$.

In the case where $\Delta_s^2 < \epsilon$:

$$(\epsilon+\Delta_s^2)^{\frac{3}{2}} \geq \epsilon^{\frac{3}{2}} \implies \frac{\epsilon|\Delta_s|^{2\alpha}}{(\epsilon+\Delta_s^2)^{\frac{3}{2}}} \leq \epsilon^{-\frac{1}{2}}|\Delta_s|^{2\alpha} < \epsilon^{\alpha-\frac{1}{2}}$$

In both cases, in the regime $\frac{1}{2} < \alpha < 1$, the integrand of (III) is bounded by $C\epsilon^{\alpha-\frac{1}{2}}$. Thus,

$$\mathbb{E}[\phi_\epsilon(\Delta_t)] \leq \mathbb{E}[\phi_\epsilon(\Delta_0)] + C_1\int_0^t \mathbb{E}[|\Delta_s|]\mathrm{d}s + C_3\epsilon^{\alpha-\frac{1}{2}}t$$

Taking the limit as $\epsilon \to 0$ by uniform convergence on the ϕ_ϵ functions yields us

$$\mathbb{E}[|\Delta_t|] \leq \mathbb{E}[|\Delta_0|] + C_1\int_0^t \mathbb{E}[|\Delta_s|]\mathrm{d}s$$

By Gronwall's inequality, for some constant C_4,

$$\mathbb{E}[|\Delta_t|] \leq C_4 e^{C_1 t}\mathbb{E}[|\Delta_0|] \leq 0$$

So $\mathbb{E}[|X_t^1 - X_t^2|] = 0\ \forall t$, which implies by continuity of solutions that $X_t^1 = X_t^2$ almost surely on an interval $[0,T]$. $\square$

5 Derive the moments $\mathbb{E}[X_t^p]$, for p $\in \mathbb{N}$, of the solution to the SDE

$$dX_t = \sqrt{X_t}dB_t, X_0 = x_0 > 0$$

Begin by considering Ito's formula applied to $f(x) = x^p$:

$$dX_t^p = pX_t^{p-1}dX_t + \frac{1}{2}p(p-1)X_t^{p-2}dQV(X_t)$$

Recall that the quadratic variance of an Ito process is calculated using only its diffusion coefficient, $QV(X_t) = \int_0^t \sigma_s^2 \mathrm{d}s$, so $dQV(X_t) = X_t \mathrm{d}t$ in our case. Then, by substituting and taking expectation:

$$\begin{aligned}
dX_t^p &= pX_t^{p-\frac{1}{2}}dB_t + \frac{1}{2}p(p-1)X_t^{p-1}dt \\
X_t^p &= x_0^p + \frac{1}{2}p(p-1)\int_0^t X_s^{p-1}\mathrm{d}s + p\int_0^t X_s^{p-\frac{1}{2}}\mathrm{d}B_s \\
\mathbb{E}[X_t^p] &= x_0^p + \frac{1}{2}p(p-1)\int_0^t \mathbb{E}[X_s^{p-1}]\mathrm{d}s + p\,\underbrace{\mathbb{E}\left[\int_0^t X_s^{p-\frac{1}{2}}\mathrm{d}B_s\right]}_{\text{(I)}}
\end{aligned}$$

where (I) vanishes under expectation assuming the integrand is L^2/the expectation exists, because it is a martingale. Observe that we can build a system of linear ODEs through the relation

$$\frac{d\mathbb{E}[X_t^p]}{dt} = \frac{1}{2}p(p-1)\mathbb{E}[X_t^{p-1}]$$

Thus, let us define the system

$$\begin{aligned}
g_p(t) &:= \mathbb{E}[X_t^p] \\
g_p'(t) &= \frac{1}{2}p(p-1)g_{p-1}(t) \\
g_p(0) &= x_0^p
\end{aligned}$$

Starting from $\mathbb{E}[X_t] = g_1(t) = x_0$,

$$\begin{aligned}
g_2'(t) = g_1(t) = x_0 &\implies g_2(t) = x_0^2 + x_0^t \\
g_3'(t) = 3(x_0^2 + x_0^t) &\implies g_3(t) = x_0^3 + 3x_0^2 t + \frac{3}{2}x_0 t^2
\end{aligned}$$

So the general form has

$$\begin{aligned}
g_p(t) &= \sum_{k=0}^{p-1} c_{p,k} x_0^{p-k} t^k \\
g_p'(t) &= \sum_{k=0}^{p-1} k c_{p,k} x_0^{p-k} t^{k-1} \qquad (**)
\end{aligned}$$

But we also know that

$$g_p'(t) = \frac{1}{2}p(p-1)g_{p-1}(t) = \sum_{j=0}^{p-2} \frac{1}{2}p(p-1)c_{p-1,j}x_0^{p-1-j}t^j \qquad (***)$$

We need to quickly deal with the summation index mismatch between k, j–in (**), the exponents t^{k-1} for $k = 0, 1, ...p-1$ are $t^0, ...t^{p-2}$, and thus, in (***), t^j for $j = 0, ...p-2$ requires an index shift $j \mapsto k-1$. Matching the per-term coefficients, we have

$$kc_{p,k}x_0^{p-k} = \frac{1}{2}p(p-1)c_{p-1,k-1}x_o^{p-1-k+1} \implies c_{p,k} = \frac{p(p-1)}{2k}c_{p-1,k-1}, \qquad c_{p,0} = 1 \ \forall p$$

We can thus construct an explicit formula for $c_{p,k}$:

$$\begin{aligned} c_{p,k} &= \frac{p(p-1)}{2k} \cdot \frac{(p-1)(p-2)}{2k-1} \cdot ... \cdot \frac{(p-k+1)(p-k)}{2} \\ &= \frac{1}{2^k k!} p(p-1)^2...(p-k+1)^2(p-k) \\ &= \frac{1}{2^k k!} \frac{p!}{(p-k)!} \frac{(p-1)!}{(p-k-1)!} \\ &= \frac{1}{2^k} \binom{p}{k} \frac{(p-1)!}{(p-k-1)!} \end{aligned}$$

and finally, plugging it into $g_p(t)$,

$$\mathbb{E}[X_t^p] = \sum_{k=0}^{p-1} x_0^{p-k} \left(\frac{t}{2}\right)^k \binom{p}{k} \frac{(p-1)!}{(p-k-1)!}$$

as desired.

6 Suppose that $X_t \in \mathbb{R}^d$ solves the SDE

$$dX_t = \sigma(t, X_t)dB_t + b(t, X_t)dt$$

where $\sigma(t, X_t)$ is a Lipschitz matrix in the second argument for an m-dimensional Brownian motion, and $b(t, X_t)$ is a Lipschitz vector in the second argument. Let A be a closed set in $\mathbb{R}^d$, and define the hitting time

$$\tau = \inf\{t \geq 0, X_t \in A\}$$

Let $f(X_t)$ be a function of only X_t, and $\mathcal{L}$ be the infinitesimal generator. Suppose that $\mathcal{L}f = \lambda f$ for $x \in A^C$, $\lambda > 0$. Prove the inequality

$$\frac{f(x_0)}{\sup_{y \in A} f(y)} \leq \mathbb{E}[e^{-\lambda\tau}|X_0 = x_0] \leq \frac{f(x_0)}{\inf_{y \in A} f(y)}$$

The intuition here is to find something that when Ito's lemma is applied to it, will make use of the eigen-identity and cancel out the generator $\mathcal{L}f$. Consider applying Ito's lemma to $e^{-\lambda t}f(X_t)$, for $t < \tau$:

$$\begin{aligned} d(e^{-\lambda t}f(X_t)) &= e^{-\lambda t}df(X_t) + f(X_t)d(e^{-\lambda t}) + \underbrace{(df(X_t))(d(e^{-\lambda t}))}_{d(e^{-\lambda t}) \text{ contributes a } dt \text{ term}} \\ &= e^{-\lambda t}(\partial_t + \mathcal{L})f(X_t)dt - \lambda e^{-\lambda t}f(X_t)dt + e^{-\lambda t}\nabla_x f(X_t)^T \sigma(t, X_t)dB_t \end{aligned}$$

Notice that the last term in the first line vanishes because a dt multiplier results in higher order differentials that all vanish. Now, applying $\mathcal{L}f = \lambda f$, and realizing that $\partial_t f(X_t) = 0$,

$$d(e^{-\lambda t}f(X_t)) = e^{-\lambda t}\nabla_x f(X_t)^T \sigma(t, X_t)dB_t$$

It is convenient to note now that the process $M_t := e^{-\lambda(t\wedge\tau)}f(X_{t\wedge\tau})$ has no drift so it is a martingale. Thus, by the optional stopping theorem $\mathbb{E}[M_\tau|x_0] = \mathbb{E}[M_0|x_0] = f(x_0)$. Letting $t \to \infty$, $(t \wedge \tau) \to \tau$ almost surely, so

$$f(x_0) = \mathbb{E}[e^{-\lambda\tau}f(X_\tau)|x_0]$$

By definition of τ, $X_\tau \in A$, so

$$\inf_{y\in A} f(y) \leq f(X_\tau) \leq \sup_{y\in A} f(y) \iff \mathbb{E}[e^{-\lambda\tau}|x_0] \inf_y f(y) \leq \underbrace{\mathbb{E}[e^{-\lambda\tau}f(X_\tau)|x_0]}_{f(x_0)} \leq \mathbb{E}[e^{-\lambda\tau}|x_0] \sup_y f(y)$$

Rearranging gives us the desired inequality: $\frac{f(x_0)}{\sup_{y\in A} f(y)} \leq \mathbb{E}[e^{-\lambda\tau}|X_0 = x_0] \leq \frac{f(x_0)}{\inf_{y\in A} f(y)}$. $\square$

7 The Stratonovich integral is defined as:

$$\int_0^t \sigma(s, B_s) \circ \mathbf{d}B_s := \lim_{|\Pi|\downarrow 0} \sum_{i=0}^{n} \frac{1}{2}[\sigma(t_i, B_{t_i}) + \sigma(t_{i+1}, B_{t_{i+1}})](B_{t_{i+1}} - B_{t_i})$$

1. **Find the transformation from the Stratonovich integral to the Ito integral, and the sufficient condition on σ**

2. **Suppose that**

$$dX_t = b(t, X_t)dt + \sigma(t, X_t) \circ dB_t$$

Find $\tilde{\sigma}, \tilde{b}$ so that

$$dX_t = \tilde{b}(t, X_t)dt + \tilde{\sigma}(t, X_t)dB_t$$

and the sufficient conditions on σ, b for this transformation

3. **Find the Radon-Nikodym derivative between the measures induced by the Stratonovich SDE and its corresponding Ito SDE,**

$$\begin{aligned} dX_t &= b(t, X_t)dt + \sigma(t, X_t) \circ dB_t \\ dX_t &= b(t, X_t)dt + \sigma(t, X_t)dB_t \end{aligned}$$

and a sufficient condition for absolute continuity on bounded time intervals

Part 1: Start by rewriting the integral as

$$\begin{aligned} \int_0^t \sigma(s, B_s) \circ \mathrm{d}B_s &= \lim_{|\Pi|\downarrow 0} \sum_{i=0}^{n} \frac{1}{2}[\sigma(t_i, B_{t_i}) + \sigma(t_{i+1}, B_{t_{i+1}})](B_{t_{i+1}} - B_{t_i}) \\ &= \lim_{|\Pi|\downarrow 0} \sum_{i=0}^{n} (B_{t_{i+1}} - B_{t_i}) \frac{1}{2}[\sigma(t_{i+1}, B_{t_{i+1}}) - \sigma(t_i, B_{t_i})] + \sum_{i=0}^{n} \sigma(t_i, B_{t_i})(B_{t_{i+1}} - B_{t_i}) \end{aligned}$$

and notice that the second term $\to \int_0^t \sigma \mathrm{d}B_s$ in Ito sense. The first term needs a bit more work:

$$\begin{aligned} &\lim_{|\Pi|\downarrow 0} \sum_{i=0}^{n} \frac{1}{2}(B_{t_{i+1}} - B_{t_i})[\sigma(t_{i+1}, B_{t_{i+1}}) - \sigma(t_i, B_{t_i})] \\ &= \lim_{|\Pi|\downarrow 0} \sum_{i=0}^{n} \frac{1}{2}(B_{t_{i+1}} - B_{t_i})^2 \underbrace{\left(\frac{\sigma(t_{i+1}, B_{t_{i+1}}) - \sigma(t_i, B_{t_i})}{B_{t_{i+1}} - B_{t_i}} \right)}_{\text{secant}} \\ &= \lim_{|\Pi|\downarrow 0} \sum_{i=0}^{n} \frac{1}{2}(B_{t_{i+1}} - B_{t_i})^2 \left(\frac{\sigma(t_{i+1}, B_{t_{i+1}}) - \sigma(t_i, B_{t_i})}{B_{t_{i+1}} - B_{t_i}} \right) \\ &= \lim_{|\Pi|\downarrow 0} \sum_{i=0}^{n} \frac{1}{2}(B_{t_{i+1}} - B_{t_i})^2 \left(\frac{\sigma(t_{i+1}, B_{t_{i+1}}) - \sigma(t_{i+1}, B_{t_i}) + \sigma(t_{i+1}, B_{t_i}) - \sigma(t_i, B_{t_i})}{B_{t_{i+1}} - B_{t_i}} \right) \\ &= \lim_{|\Pi|\downarrow 0} \underbrace{\sum_{i=0}^{n} \frac{1}{2}(B_{t_{i+1}} - B_{t_i}) \left(\sigma(t_{i+1}, B_{t_i}) - \sigma(t_i, B_{t_i})\right)}_{(I)} + \underbrace{\sum_{i=0}^{n} \frac{1}{2}(B_{t_{i+1}} - B_{t_i})^2 \left(\frac{\sigma(t_{i+1}, B_{t_{i+1}}) - \sigma(t_{i+1}, B_{t_i})}{B_{t_{i+1}} - B_{t_i}} \right)}_{(II)} \end{aligned}$$

Applying mean value theorem in t,

$$(I) = \frac{1}{2} \sum_{i=0}^{n} \sigma_t(\gamma_i, B_{t_i})(B_{t_{i+1}} - B_{t_i})(t_{i+1} - t_i) \to \int_0^t \sigma_s(s, B_s) \mathrm{d}s \mathrm{d}B_s = 0$$

for some $\gamma_i \in (t_i, t_{i+1})$, with convergence as $n \to \infty$.

To be more precise, $\sigma_t(\gamma_i, B_{t_i})$ is adapted to the standard Brownian/Ito filtration, and because Brownian paths are uniformly continuous on compact sets,

$$\sum_j (B_{t_{j+1}} - B_{t_j})(t_{j+1} - t_j) \leq \max_j |B_{t_{j+1}} - B_{t_j}| \sum_j (t_{j+1} - t_j) \to 0$$

as the mesh size $|\Pi| \downarrow 0$ (also the same reason why $dtdB_t$ higher order differentials vanish).

Note that implicitly here, we are assuming $\sigma \in C_t^1$ through mean value theorem and the continuity of the integrand $\gamma_i \mapsto t$. (Also note that we could have handled (I) by citing continuity of σ in t and treating it as the limit of difference of two $\int_0^t \sigma \mathrm{d}B_s$ integrals, which requires less regularity on σ). To deal with (II), define $\Delta B_{t_i} := B_{t_{i+1}} - B_{t_i}$, and

$$\begin{aligned}
(II) &= \frac{1}{2} \sum_i \left(\frac{\sigma(t_{i+1}, B_{t_{i+1}}) - \sigma(t_{i+1}, B_{t_i})}{\Delta B_{t_i}} \right) (\Delta B_{t_i})^2 \\
&= \frac{1}{2} \sum_i \sigma_x(t_{i+1}, \xi_i)(\Delta B_{t_i})^2 \\
&= \frac{1}{2} \left(\sum_i \sigma_x(t_{i+1}, \xi_i)(t_{i+1} - t_i) + \sum_i \sigma_x(t_{i+1}, \xi_i)((\Delta B_{t_i})^2 - (t_{i+1} - t_i)) \right)
\end{aligned}$$

again justified by using mean value theorem on σ in its second argument, for some $\xi_i \in (B_{t_i}, B_{t_{i+1}})$. Assuming the condition $\sigma \in C_x^1$ on a compact set implies that it is bounded, and recall that the variance of $(\Delta B_i)^2$ is equal to $t_{i+1} - t_i$ from the derivation of Brownian motion's quadratic variance. As $n \to \infty$, we have

$$\sum_i \sigma_x(t_{i+1}, \xi_i)((\Delta B_{t_i})^2 - (t_{i+1} - t_i)) \leq \sup_i \sigma_x(t_{i+1}, \xi_i) \sum_i (\Delta B_{t_i})^2 - (t_{i+1} - t_i) \to \underbrace{0}_{\text{expectation}}$$

$$\frac{1}{2} \sum_i \sigma_x(t_{i+1}, \xi_i)(t_{i+1} - t_i) \to \frac{1}{2} \int_0^t \sigma_x(s, B_s)\mathrm{d}s$$

where again, assuming continuity in the second argument makes $\xi_i \mapsto B_{t_i}$ justifiable as we integrate ds. Thus, the Stratonovich integral converges to:

$$\int_0^t \sigma(s, B_s)\mathrm{d}B_s + \frac{1}{2} \int_0^t \sigma_x(s, B_s)\mathrm{d}s$$

with sufficient assumption being $\sigma \in C_{t,x}^{1,1}$. But can we do better?

Recall the initial sum that we split up:

$$\lim_{|\Pi| \downarrow 0} \sum_{i=0}^n (B_{t_{i+1}} - B_{t_i}) \frac{1}{2} [\sigma(t_{i+1}, B_{t_{i+1}}) - \sigma(t_i, B_{t_i})] + \sum_{i=0}^n \sigma(t_i, B_{t_i})(B_{t_{i+1}} - B_{t_i}) \to \int_0^t \sigma(s, B_s)\mathrm{d}B_s + \frac{1}{2} CV(\sigma, B_t)$$

where $CV(\sigma, B_t)$ is the cross variation between σ, B_t. To find this covariation directly, examine σ in Ito differential form:

$$d\sigma(t, B_t) = \sigma_x(t, B_t)dB_t + \left(\sigma_t(t, B_t) + \frac{1}{2}\sigma_{tt}(t, B_t) \right) dt$$

Because cross variation is essentially a $d\sigma(t, X_t)dB_t$ cross term in integral form, its contribution will be only coming from the diffusion terms. In other words,

$$d\sigma(t, B_t)dB_t = \sigma_x(t, B_t)(dB_t)^2 + \left(\sigma_t(t, B_t) + \frac{1}{2}\sigma_{tt}(t, B_t) \right) (dt)(dB_t)$$

and it is obvious that cross terms ($dtdB_t$) vanish, and that $(dB_t)^2 = dt$. Thus,

$$CV(\sigma, B_t) = \int_0^t \mathrm{d}\sigma(s, B_s)\mathrm{d}B_s = \int_0^t \sigma_x(s, B_s)\mathrm{d}s$$

and the Stratonovich conversion is

$$\int_0^t \sigma(s, B_s) \circ \mathrm{d}B_s = \int_0^t \sigma(s, B_s)\mathrm{d}B_s + \frac{1}{2}\int_0^t \sigma_x(s, B_s)\mathrm{d}s$$

which matches the result we got before. But for this derivation, the sufficient condition is only that σ, σ_x are "Brownian-integrable" on $[0, t]$, i.e. $\sigma, \sigma_x \in L^2_{loc}$, and that σ_{xx}, σ_t exist for the use of Ito's lemma (this condition is traded for $\sigma \in C^1_x$ from before).

Part 2: Consider the stratonovich SDE in integral form:

$$X_t = X_0 + \int_0^t b(s, X_s)\mathrm{d}s + \underbrace{\int_0^t \sigma(s, X_s) \circ \mathrm{d}B_s}_{(*)}$$

so the question reduces to–what is the Stratonovich integral (*)? Similar to part 1, let us write:

$$\begin{aligned}
(*) &= \lim_{|\Pi|\downarrow 0} \sum_{i=0}^{n-1} \left(\frac{\sigma(t_{i+1}, X_{t_{i+1}}) - \sigma(t_i, X_{t_i})}{2}\right)(B_{t_{i+1}} - B_{t_i}) + \sum_{i=0}^{n-1}(B_{t_{i+1}} - B_{t_i})\sigma(t_i, X_{t_i}) \\
&= \frac{1}{2}CV(\sigma(t, X_t), B_t) + \int_0^t \sigma(s, X_s)\mathrm{d}B_s
\end{aligned}$$

As before, apply Ito's lemma to $\sigma(t, X_t)$:

$$d\sigma(t, X_t) = \sigma_x(t, X_t)dX_t + \sigma_t(t, X_t)dt + \frac{1}{2}\sigma_{xx}(t, X_t)dQV(X_t)$$

Assuming it exists (reasonable considering the question), the Ito formulation of X_t is $dX_t = \tilde{b}(t, X_t)dt + \tilde{\sigma}(t, X_t)dB_t$, and because X_t is an Ito process, $dQV(X_t) = \tilde{\sigma}^2(t, X_t)dt$. Thus, the differential form of cross variation is

$$\begin{aligned}
d\sigma(t, X_t)dB_t &= \sigma_x(t, X_t)(\tilde{b}(t, X_t)dt + \tilde{\sigma}(t, X_t)dB_t)dB_t + \left(\sigma_t(t, X_t) + \frac{1}{2}\sigma_{xx}(t, X_t)\tilde{\sigma}^2(t, X_t)\right)dtdB_t \\
&= \sigma_x(t, X_t)\tilde{\sigma}(t, X_t)dt
\end{aligned}$$

as higher order cross terms all vanish. To determine what $\tilde{\sigma}$ is, consider that:

$$\begin{aligned}
\int_0^t \sigma(s, X_s) \circ \mathrm{d}B_s &= \int_0^t \sigma(s, X_s)\mathrm{d}B_s + \frac{1}{2}\int_0^t \mathrm{d}\sigma(s, X_s)\mathrm{d}B_s \\
&= \int_0^t \sigma(s, X_s)\mathrm{d}B_s + \frac{1}{2}\int_0^t \sigma_x(s, X_s)\tilde{\sigma}(s, X_s)\mathrm{d}s
\end{aligned}$$

and so the differential SDE of X_t is:

$$dX_t = \left(b(t, X_t) + \frac{\sigma_x(t, X_t)\tilde{\sigma}(t, X_t)}{2}\right)dt + \sigma(t, X_t)dB_t$$

which is written purely in Ito form. It only remains to match coefficients, which means $\tilde{\sigma}(t, X_t) = \sigma(t, X_t)$, $\tilde{b}(t, X_t) = b(t, X_t) + \frac{1}{2}\sigma_x(t, X_t)\sigma(t, X_t)$. The sufficient assumptions in this case are that $\sigma_t, \sigma_x, \sigma_{xx}$ should exist for Ito's lemma, and $b, \sigma, \sigma_x\sigma \in L^2_{loc}$ for integration. Furthermore, we also want existence and uniqueness of X_t, so σ, b being 1-Lipschitz or σ being α-Holder is required.

Part 3: Define the path measure induced by an Ito SDE $dX_t = \sigma dB_t + bdt$ to be $\mathbb{P}_x^{\sigma^2,b}$ on $\mathcal{C}([0,T])$. For absolute continuity and existence of the Radon-Nikodym derivative $\frac{d\mathbb{P}_2}{d\mathbb{P}_1}$, we appeal to Girsanov's theorem. Since the Stratonovich SDE takes the Ito form $dX_t = \sigma dB_t + (b + \frac{1}{2}\sigma_x\sigma)dt$, its induced measure is $\mathbb{P}_x^{\sigma^2,(b+\frac{1}{2}\sigma_x\sigma)}$. Girsanov's change of measure gives us the R-N derivative:

$$\begin{aligned}\frac{d\mathbb{P}_x^{\sigma^2,(b+\frac{1}{2}\sigma_x\sigma)}}{d\mathbb{P}_x^{\sigma^2,b}} &= e^{-\int_0^t \frac{1}{\sigma}(b+\frac{1}{2}\sigma_x\sigma - b)\mathrm{d}B_s - \frac{1}{2}\int_0^t \frac{1}{\sigma^2}(b+\frac{1}{2}\sigma_x\sigma-b)^2 \mathrm{d}s} \\ &= e^{-\frac{1}{2}\int_0^t \sigma_x \mathrm{d}B_s - \frac{1}{8}\int_0^t \sigma_x^2 \mathrm{d}s}\end{aligned}$$

for which a sufficient condition is Novikov's condition (which ensures that the R-N derivative is a martingale), given by:

$$\mathbb{E}[e^{\frac{1}{8}\int_0^t \sigma_x^2 \mathrm{d}s}] < \infty$$

The condition also guarantees absolute continuity on bounded time intervals, and all conditions from parts 1 & 2 are required for this to make sense.

8 Let $X_t \in \mathbb{R}^d$ solve the SDE

$$dX_t = \sigma(t, X_t)dB_t + b(t, X_t)dt$$

where $\sigma(t, X_t)$ is a Lipschitz matrix in the second argument for $B_t \in \mathbb{R}^m$, and $b(t, X_t)$ is a Lipschitz vector in the second argument. Suppose $f \in C^{1,2}_{t,x}$. Find a function $g(t, X_t), g(0, X_0) = 1$ where the product $f(t, X_t)g(t, X_t)$ is a martingale.

Recall that an Ito process is a martingale when it is drift-less, i.e. $Y_t = \tilde{\sigma}(t, Y_t)dB_t$, assuming nice enough conditions. We seek to make $Y_t := f(t, X_t)g(t, X_t)$ drift-less. First, apply Ito's formula on $f(t, X_t)$:

$$\begin{aligned}df(t, X_t) &= (\partial_t + \mathcal{L})f(t, X_t)dt + \nabla_x f(t, X_t)^T \sigma(t, X_t)dB_t \\ \mathcal{L} &= \underbrace{\frac{1}{2}\sum_{i,j=1}^d a_{ij}\frac{\partial^2}{\partial x_i \partial x_j}} + \sum_{i=1}^d b_i \frac{\partial}{\partial x_i} \\ \mathcal{L}f &= \frac{1}{2}\mathrm{tr}(\sigma\sigma^T \nabla_x^2 f) + b \cdot \nabla_x f \\ a(t, X_t) &= \sigma(t, X_t)\sigma(t, X_t)^T\end{aligned}$$

and we can identify the drift and diffusion coefficients easily from the first line. Define them as:

$$\begin{aligned}A(t, X_t) &:= \partial_t f(t, X_t) + \mathcal{L}f(t, X_t) \\ H(t, X_t) &:= \sigma(t, X_t)^T \nabla_x f(t, X_t)\end{aligned}$$

Our best guess here for the form of $g(t, X_t)$ is that it will be a martingale, and when Ito's lemma is applied to it, it will help cancel the drift in the intermediate result. So we should let

$$dg(t, X_t) = g(t, X_t)\theta(t, X_t)^T dB_t$$

Then, Ito's formula on fg yields (omitting (t, X_t) dependence in notation):

$$\begin{aligned}d(f(t, X_t)g(t, X_t)) &= fdg + gdf + (df)(dg) \\ &= fg\theta^T dB_t + gAdt + gH^T dB_t + (Adt + H^T dB_t)(g\theta^T dB_t) \\ &= gAdt + gH^T\theta dt + gH^T dB_t + fg\theta^T dB_t\end{aligned}$$

where the $dtdB_t$ factor vanishes due to it being a higher order stochastic differential. For the martingale property, it is sufficient that

$$g(t,X_t)A(t,X_t) + g(t,X_t)H(t,X_t)^T\theta(t,X_t) = 0 \impliedby A(t,X_t) + H(t,X_t)^T\theta(t,X_t) = 0$$

which (the latter) is a linear equation in the unknown $\theta(t,X_t)$. Since $H, \theta \in \mathbb{R}^m$ are one-dimensional, we choose the canonical solution $\theta = \lambda H$, so that

$$\lambda(t,X_t)\|H(t,X_t)\|^2 = -A(t,X_t) \implies \theta(t,X_t) = -\frac{A(t,X_t)}{\|H(t,X_t)\|^2}H(t,X_t)$$

and we have that

$$dg(t,X_t) = -g(t,X_t)\frac{A(t,X_t)}{\|H(t,X_t)\|^2}H(t,X_t)^T dB_t, \qquad\qquad g(0,X_0) = 1$$

To solve for g, the trick is to apply Ito's lemma on $\ln(g(t,X_t))$:

$$\begin{aligned} d\ln(g(t,X_t)) &= \frac{1}{g}dg + \frac{1}{2}\left(-\frac{1}{g^2}\right)(dg)^2 \\ &= -\theta^T dB_t - \frac{1}{2}\frac{(g\theta)^2}{g^2}(dB_t)^2 \\ &= -\theta^T dB_t - \frac{1}{2}\|\theta\|^2 dt \end{aligned}$$

and in integral form,

$$\begin{aligned} \ln(g(t,X_t)) &= \ln(g(0,X_0)) - \int_0^t \theta(s,X_s)^T \mathrm{d}B_s - \frac{1}{2}\int_0^t \|\theta(s,X_s)\|^2 \mathrm{d}s \\ \implies g(t,X_t) &= e^{-\frac{1}{2}\int_0^t \|\theta(s,X_s)\|^2 \mathrm{d}s - \int_0^t \theta(s,X_s)^T \mathrm{d}B_s} \\ \theta(t,X_t) &= -\frac{(\partial_t + \mathcal{L})f(t,X_t)}{\|\sigma(t,X_t)^T\nabla_x f(t,X_t)\|^2}\nabla_x f(t,X_t)^T\sigma(t,X_t) \end{aligned}$$

is the function that makes Y_t a martingale.